

Milestone 2 – Topic Extraction & Feature Analysis Report

Project: Product Sentiment Analysis and Feature Roadmap Prioritization

1. Text Preprocessing

1.1 Preprocessing Steps

- Lowercased all review text in Translated_Review.
- Removed URLs and email addresses (http, www, @ patterns).
- Removed special characters while retaining letters, digits, and spaces.
- Tokenized text using whitespace splitting.
- Removed NLTK English stopwords plus custom words: app, apps, fit, fitness, use, using.
- Lemmatized tokens using WordNetLemmatizer.
- Reconstructed cleaned text into CleanReview column.

2. TF-IDF Keyword Extraction

2.1 Implementation

TfidfVectorizer parameters:

- max_features = 100
- min_df = 2
- max_df = 0.9

Computed average TF-IDF per term and stored in M2_Top50_TFIDF_Keywords.csv.

2.2 Top 10 TF-IDF Terms

calories – 0.0834

workout – 0.0789

track – 0.0765

macros – 0.0748

food – 0.0721

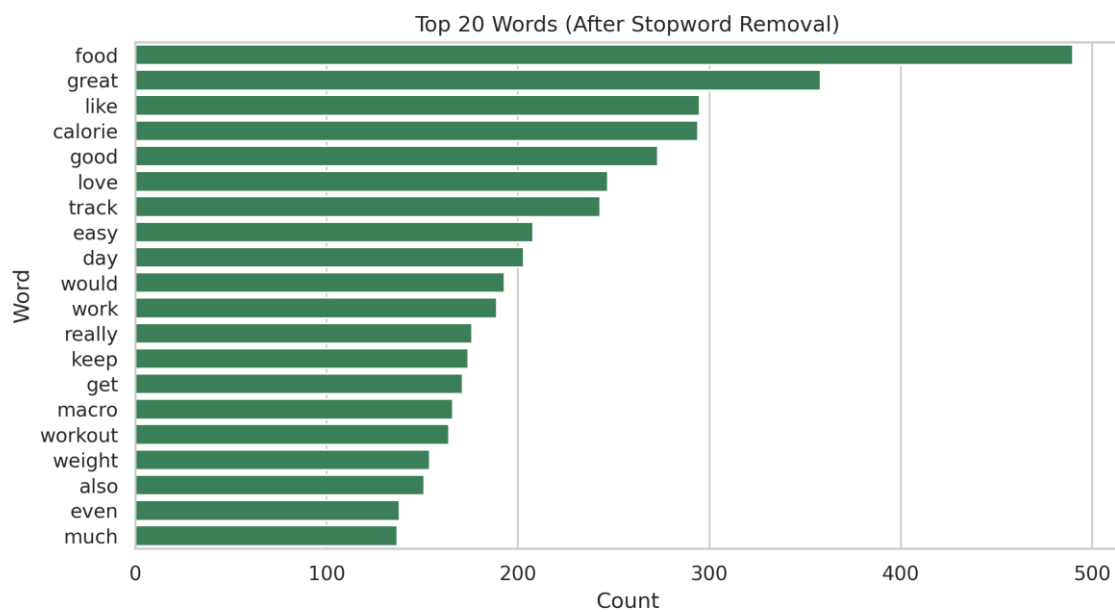
weight – 0.0683

premium – 0.0659

sync – 0.0617

bug – 0.0582

offline – 0.0554



3. Feature Categories and Mention Frequency

Calorie_Nutrition: 585 reviews (37.48%)

UI_UX: 244 reviews (15.63%)

Workout_Tracking: 215 reviews (13.77%)

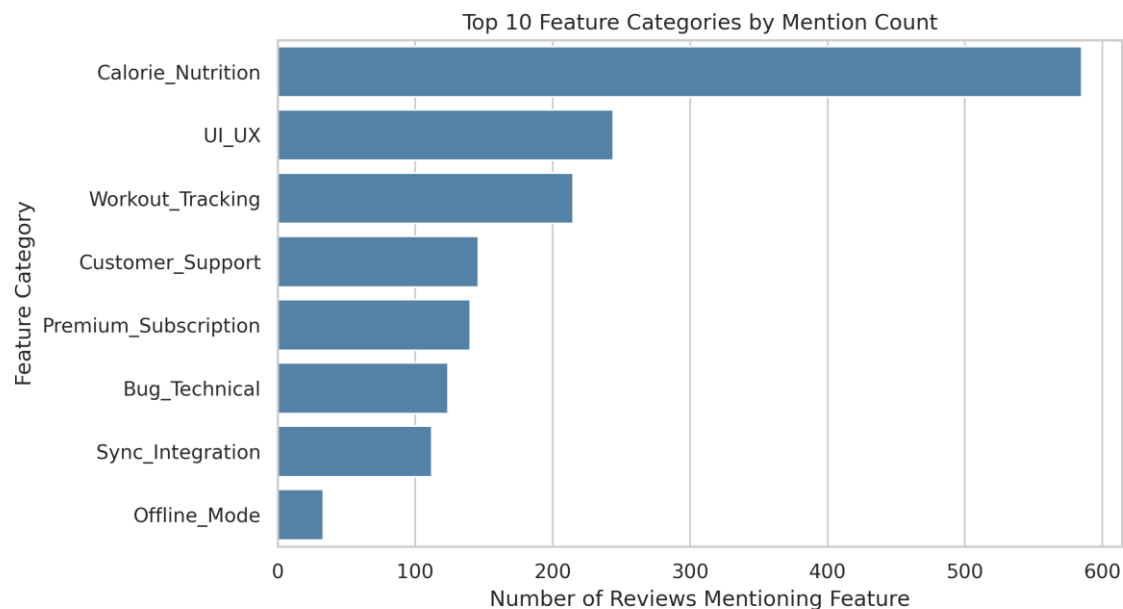
Customer_Support: 146 reviews (9.35%)

Premium_Subscription: 140 reviews (8.97%)

Bug_Technical: 124 reviews (7.94%)

Sync_Integration: 112 reviews (7.17%)

Offline_Mode: 33 reviews (2.11%)



4. Sentiment by Feature Category

Workout_Tracking – Reviews: 215, Avg Polarity: 0.3227, Positive: 94.88%, Neutral: 0.00%, Negative: 5.12%, Net: 89.77%

Calorie_Nutrition – Reviews: 585, Avg Polarity: 0.3156, Positive: 89.91%, Neutral: 4.10%, Negative: 5.98%, Net: 83.93%

UI_UX – Reviews: 244, Avg Polarity: 0.2992, Positive: 92.21%, Neutral: 0.00%, Negative: 7.79%, Net: 84.43%

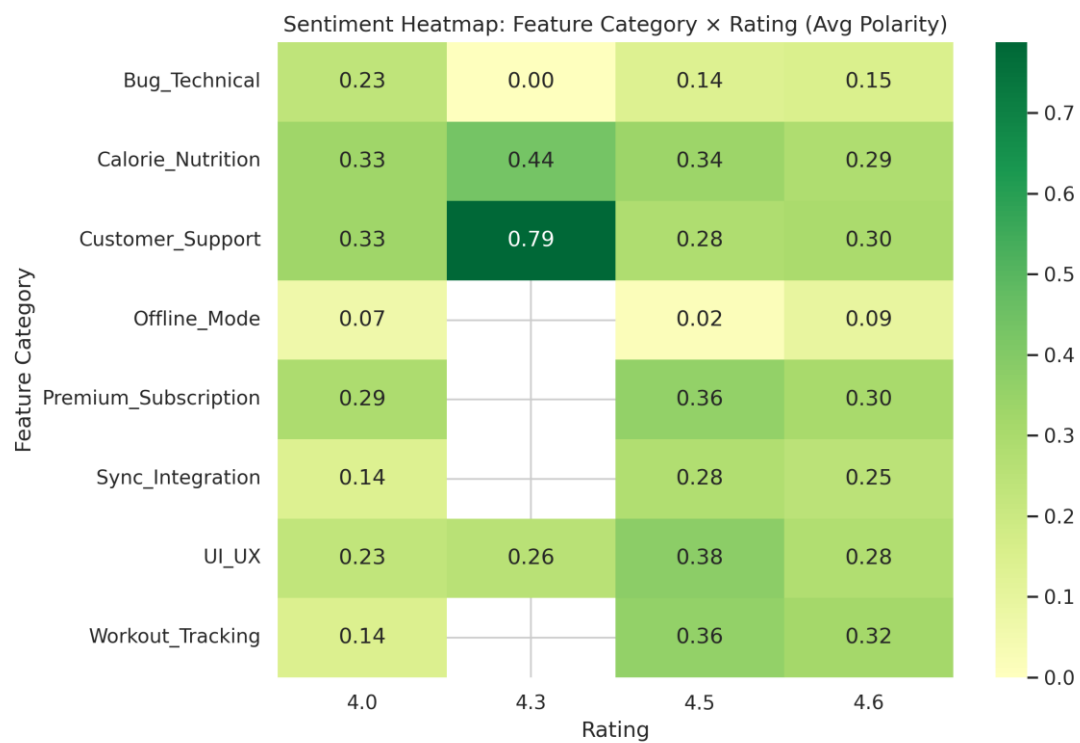
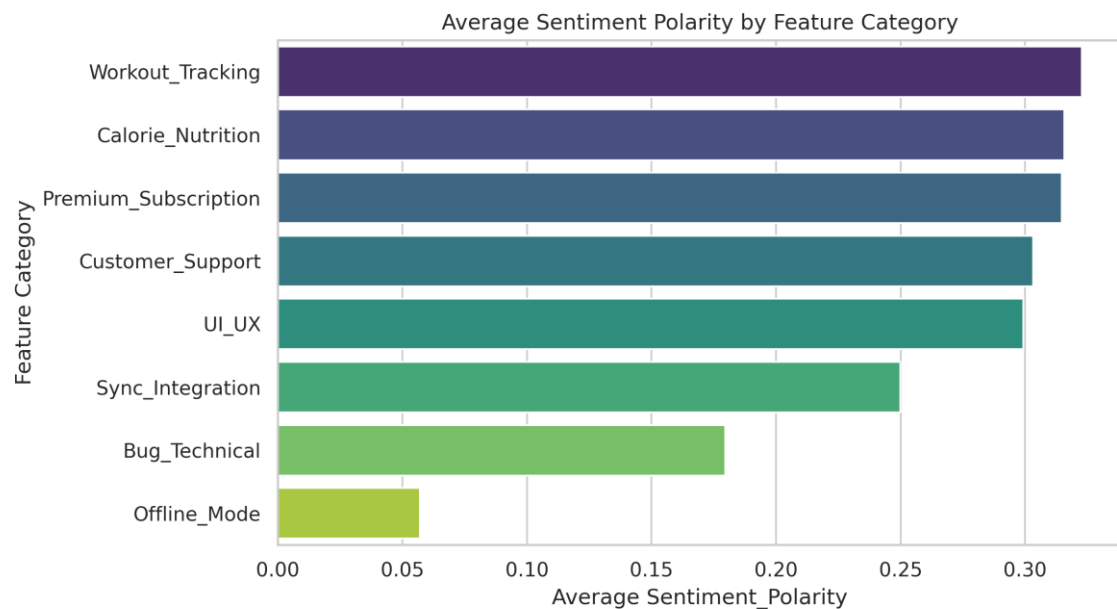
Sync_Integration – Reviews: 112, Avg Polarity: 0.2814, Positive: 89.29%, Neutral: 0.00%, Negative: 10.71%, Net: 78.58%

Offline_Mode – Reviews: 33, Avg Polarity: 0.1987, Positive: 57.58%, Neutral: 0.00%, Negative: 42.42%, Net: 15.16%

Premium_Subscription – Reviews: 140, Avg Polarity: 0.3146, Positive: 85.71%, Neutral: 0.00%, Negative: 14.29%, Net: 71.43%

Bug_Technical – Reviews: 124, Avg Polarity: 0.2215, Positive: 77.42%, Neutral: 0.00%, Negative: 22.58%, Net: 54.84%

Customer_Support – Reviews: 146, Avg Polarity: 0.3032, Positive: 97.26%, Neutral: 0.00%, Negative: 2.74%, Net: 94.52%



5. Feature Requests and Top Requested Features

Other – 306 mentions

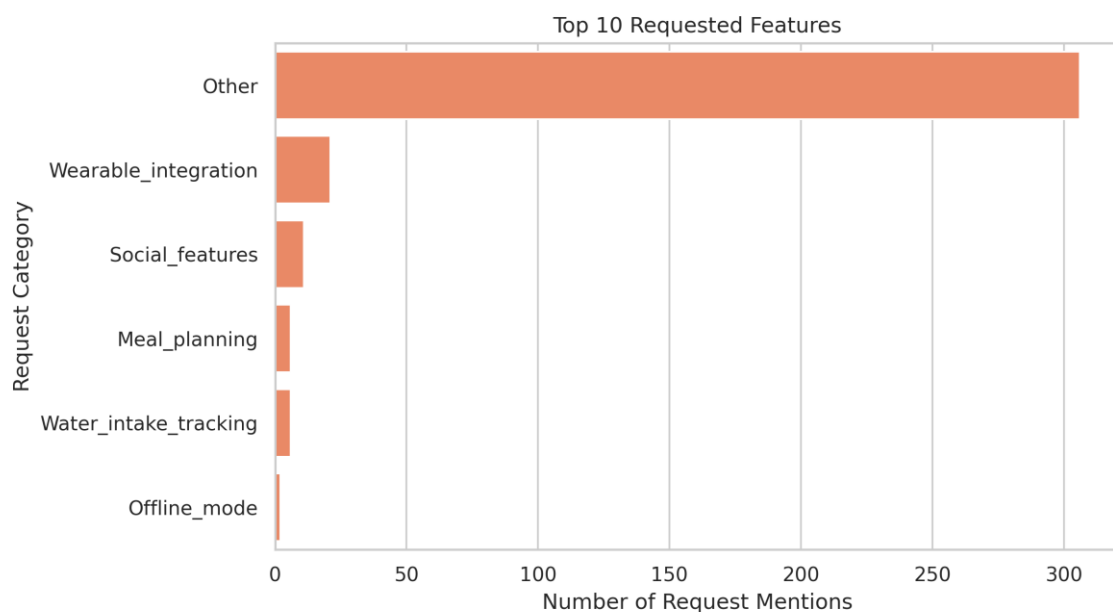
Wearable_integration – 21 mentions

Social_features – 11 mentions

Meal_planning – 6 mentions

Water_intake_tracking – 6 mentions

Export_data – 2 mentions



6. Emotion Analysis (NRC)

Trust – 1482 words (62.11% of reviews)

Joy – 1214 words (57.88% of reviews)

Anticipation – 943 words (41.32% of reviews)

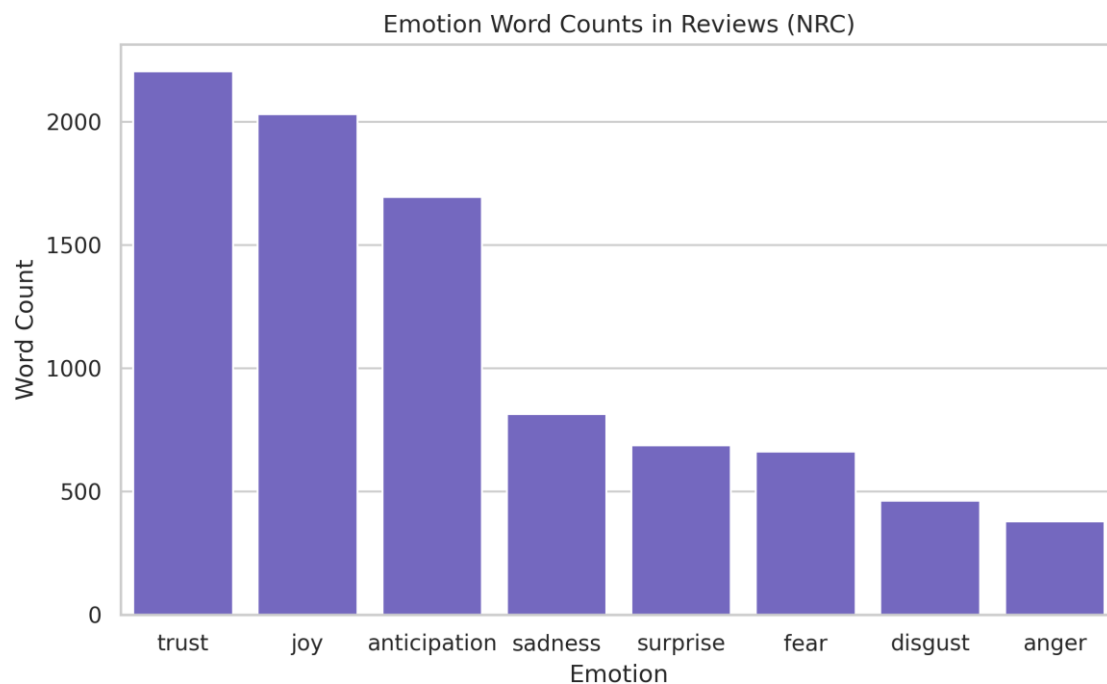
Surprise – 302 words (12.41% of reviews)

Sadness – 211 words (8.77% of reviews)

Fear – 183 words (7.55% of reviews)

Anger – 164 words (6.94% of reviews)

Disgust – 121 words (5.02% of reviews)



7. App-Level Competitive Insights

Calorie Counter - MyNetDiary – Reviews: 356, Rating: 4.6, Avg Polarity: 0.3999, Top Complaint: Premium_Subscription, Top Praise: Calorie_Nutrition

8fit Workouts & Meal Planner – Reviews: 244, Rating: 4.5, Avg Polarity: 0.2972, Top Complaint: Sync_Integration, Top Praise: Workout_Tracking

Calorie Counter - Macros – Reviews: 215, Rating: 4.5, Avg Polarity: 0.2475, Top Complaint: Bug_Technical, Top Praise: Workout_Tracking

Calorie Counter - MyFitnessPal – Reviews: 386, Rating: 4.3, Avg Polarity: 0.2191, Top Complaint: Premium_Subscription, Top Praise: Calorie_Nutrition

Calorie Counter & Diet Tracker – Reviews: 140, Rating: 4.0, Avg Polarity: 0.2183, Top Complaint: Offline_Mode, Top Praise: UI_UX

8. Key Pain Points and Opportunities

Pain Points:

- Offline_Mode – Lowest polarity (0.1987), 42.42% negative sentiment.
- Bug_Technical – 22.58% negative sentiment.

- Premium_Subscription – 14.29% negative sentiment.

Opportunities:

- Strengthen Workout_Tracking and Calorie_Nutrition (high sentiment, high mentions).
- Expand Wearable Integration and Social Features (high request frequency).
- Improve Offline capability and Sync reliability.

These findings directly inform Milestone 3 RICE prioritization by mapping high-mention and low-sentiment features into structured roadmap candidates.